



Gowin Clock User Guide

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1 About This Guide

1.1 Purpose

This manual provides a description of the clock function, definition, and usage.

It consists of four main parts:

1. Clock Resource: Description of the clock resource type, distribution, and related primitives;
2. OSC: Overview of the OSC frequency range, parameters setting, and primitives instantiation;
3. DLL: Guidance on the function and definition of DLL and DLLDLY;
4. PLL: Outline on the function and definition of PLL.

1.2 Supported Products

The information presented in this guide applies to the following Gowin FPGA products:

1. GW1N series of FPGA products: GW1N-1, GW1N-2, GW1N-4, GW1N-6, and GW1N-9.
2. GW1NR series of FPGA products: GW1NR-4 and GW1NR-9;
3. GW2A series of FPGA products: GW2A-18 and GW2A-55;
4. GW2AR series FPGA products: GW2AR-18

1.3 Related Documents

The latest user guides are available on GOWINSEMI Website. You can find the related documents at www.gowinsemi.com:

1. GW1N series of Products Data Sheet
2. GW1NR series of FPGA Products Data Sheet
3. GW1N series of FPGA Products Package and Pinout
4. GW1NR series of FPGA Products Package and Pinout
5. GW1N series of FPGA Products Programming and Configuration User Guide
6. GW2A series of FPGA Products Data Sheet
7. GW2AR series of FPGA Products Data Sheet
8. GW2A series of FPGA Products Package and Pinout
9. GW2AR series of FPGA Products Package and Pinout
10. GW1N series of FPGA Products Programming and Configuration User Guide

1.4 Abbreviations and Terminology

The abbreviations and terminologies used in this manual are set out in Table 1-1 below.

Table 1-1 Abbreviations and Terminologies

Abbreviations and Terminology	Name	Meaning
FPGA	Field Programmable Gate Array	Field Programmable Gate Array
OSC	Oscillator	Oscillator
DLL	Delay-locked Loop	DLL
DLLDLY	DLL Delay	DLL Delay
PLL	Phase-locked Loop	PLL
GCLK	Global Clock	Global Clock
HCLK	High Clock	HCLK
BUFG	Buffer Global	Buffer Global
DCS	Dynamic Clock Selector	Dynamic Clock Selector
DQCE	Dynamic Quadrant Clock Enable	Dynamic Quadrant Clock Enable
CLKDIV	Clock Divider	Clock Divider

1.5 Support and Feedback

Gowin Semiconductor provides customers with comprehensive technical support. If you have any questions, comments, or suggestions, please feel free to contact us directly using the information provided below.

Website: www.gowinsemi.com

E-mail: support@gowinsemi.com

Tel: +86 755 8262 0391

2Overview

This chapter describes the clock functionality that is available with the GOWINSEMI FPGA products, including an overview of the dedicated clock input, buffers, and routing resources. The basic clock resources include a series of low-capacity and low-offset interconnect wires that are suitable for high frequency signals and can help reduce clock jitter and improve the performance of all clock signals.

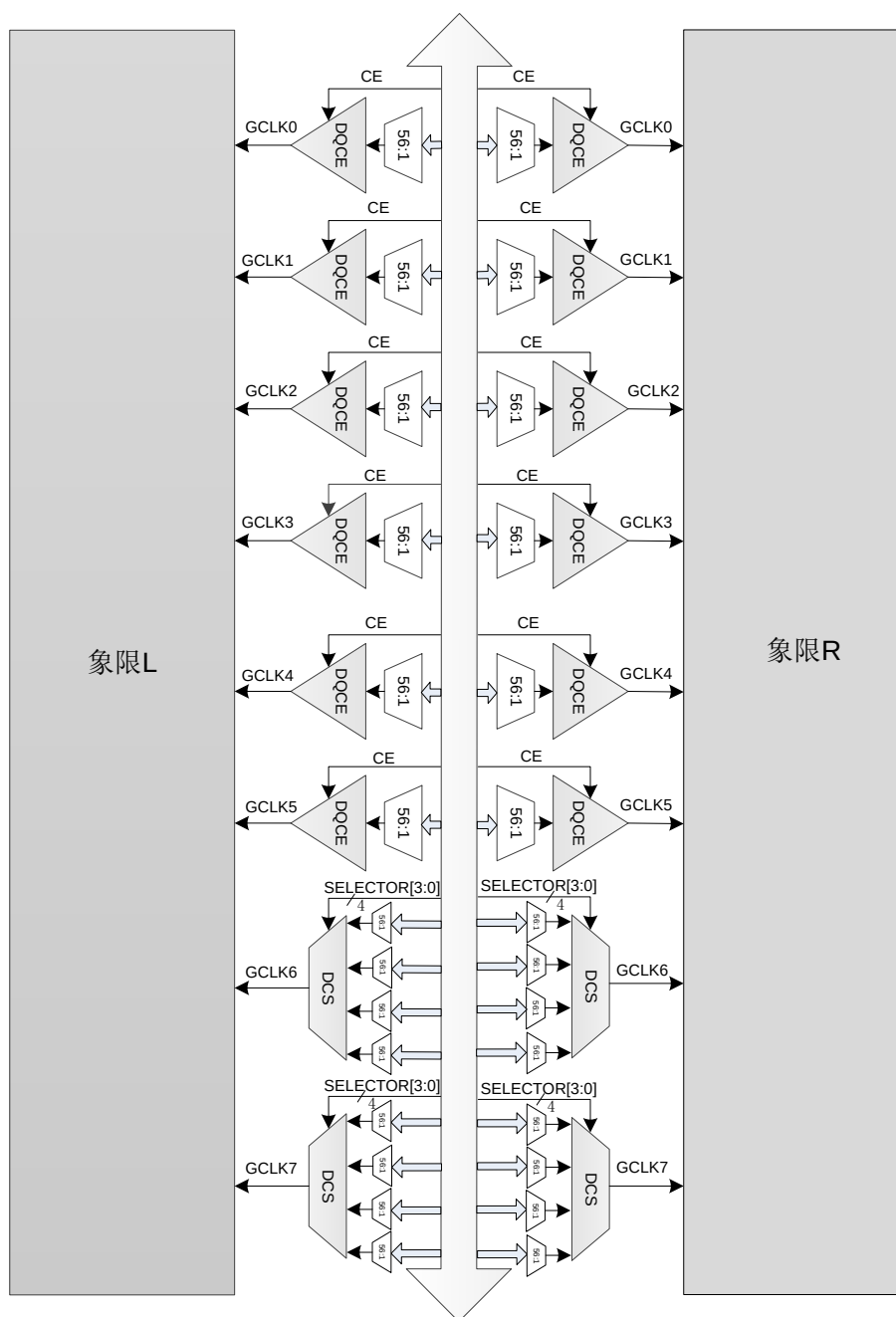
The clock resources and wiring are critical for high-performance applications in FPGA. GOWINSEMI FPGA products provide the global clock network (GCLK) which connects directly to all the registers. Besides the global clock network, the GW2A series of FPGA products provide PLL, DLL, high speed clock HCLK, DDR memory interface, and DQS, etc.

3Global Clock

3.1 Overview

GW1N/GW1NR Series of FPGA Products

The GCLK in the GW1N/GW1NR series of FPGA products distributes in the L quadrant and R quadrant, as shown in Figure 3-1. Each quadrant provides eight GCLK networks. The optional clock resources of the GCLK can be IOs or CRU. Selecting the clock from the dedicated IOs can result in better timing.

Figure 3-1 GW1N/GW1NR Series of FPGA products GCLK Quadrant Distribution**GW2A/GW2AR Series of FPGA Products**

The GCLK is distributed in GW2A/GW2AR devices as four quadrants. Each quadrant provides eight GCLKs. The optional clock resources that are included with the GCLK can be I/Os or CRU. Selecting the clock from the dedicated I/Os can result in better timing.

Figure 3-2 Clock Resources in the GW2A/GW2AR Series of FPGA Products

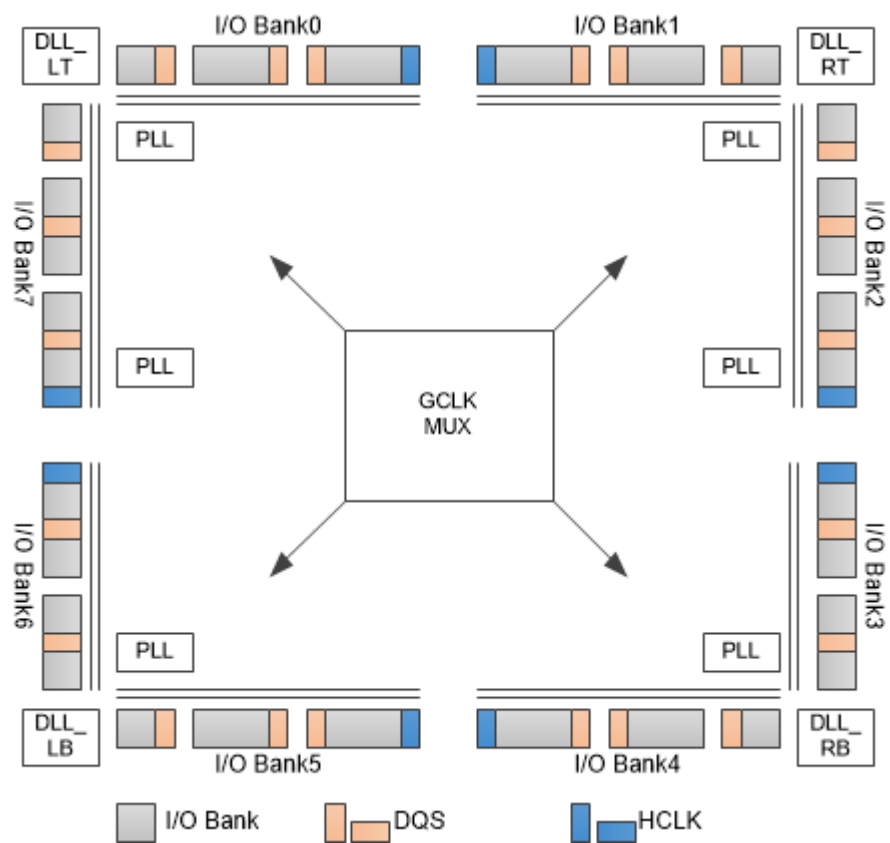
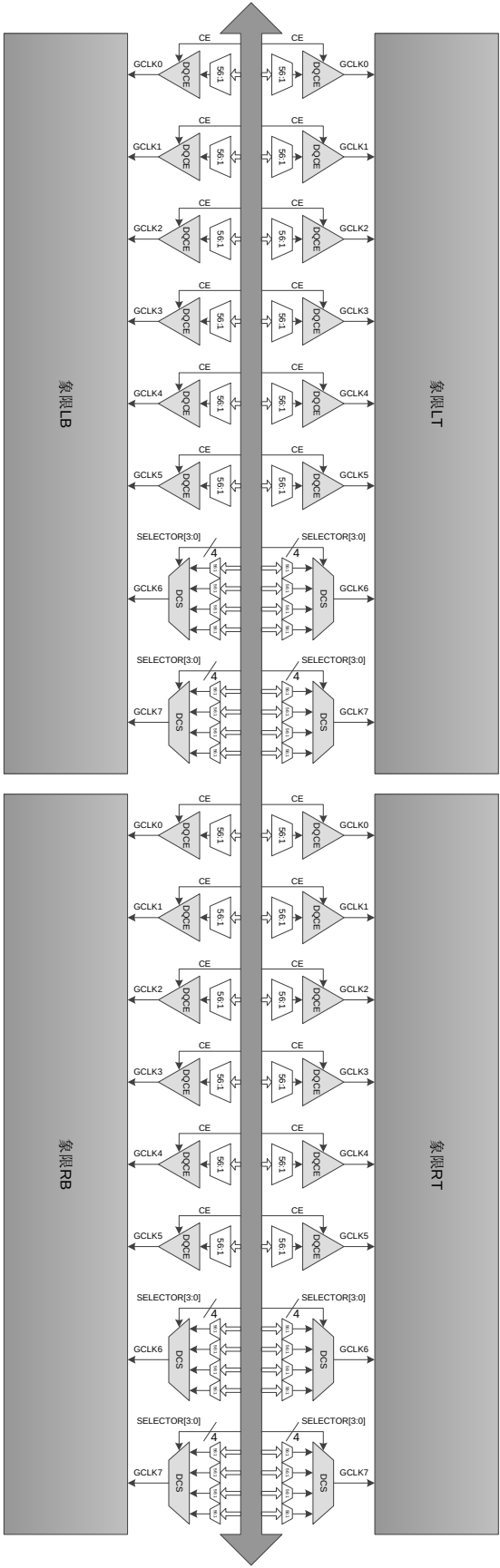


Figure 3-3 GCLK Quadrant Distribution in the GW2A/GW2AR Series of FPGA products



3.2 BUFG

3.2.1 Introduction

The BUFG buffer offers high fan-out global clocks and high fan-out driven strength. It can connect the signal to the global clock network with negligible clock jitter. BUFG components can also be used for typical high fan-out signals and networks, such as reset signals and clock enable signals. BUFG is required to implement global clock management, such as PLL, DLL, etc.

3.2.2 Primitive

VHDL

```
COMPONENT BUFG
  PORT(
    O : out std_logic;
    I : in std_logic
  );
end COMPONENT;
```

Verilog

```
module BUFG (O, I);
  output O;
  input I;
endmodule
```

3.2.3 Port

Table 3-1 BUFG Port Signal

Port	Description
I	Clock input
O	Clock output

3.3 DCS

3.3.1 Introduction

GCLK6~GCLK7 of each quadrant is controlled by the dynamic clock selector (DCS); select one of the four clocks as the global clock. Switch dynamically between CLK0~CLK3 by CRU, and output a glitch-free clock.

3.3.2 Primitive

VHDL

```
COMPONENT DCS
  GENERIC (
    DCS_MODE : string := "RISING"
```

```

);
PORT (
    CLK0 : IN std_logic;
    CLK1 : IN std_logic;
    CLK2 : IN std_logic;
    CLK3 : IN std_logic;
    CLKSEL : IN std_logic_vector(3 downto 0);
    SELFCE : IN std_logic;
    CLKOUT : OUT std_logic
);
end COMPONENT;

```

Verilog

```

module DCS (CLK0, CLK1, CLK2, CLK3, CLKSEL, SELFCE, CLKOUT);
    input CLK0, CLK1, CLK2, CLK3, SELFCE;
    input [3:0] CLKSEL;
    output CLKOUT;
    parameter DCS_MODE = "RISING";
endmodule

```

3.3.3 Port

Table 3-2 DCS Port Signal

Port	Description
CLKSEL [3:0]	Clock select signal
CLK0	Clock input 0
CLK1	Clock input 1
CLK2	Clock input 2
CLK3	Clock input 3
SELFCE	0: Glitchless mode 1: Non-glitchless mode
CLKOUT	Clock output

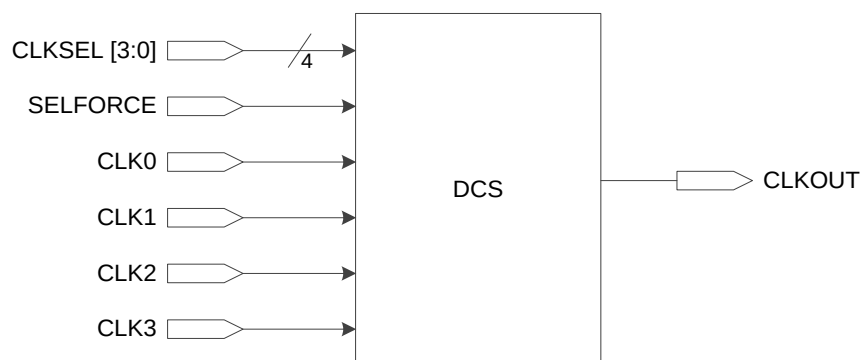
3.3.4 DCS Parameters

Table 3-3 DCS Parameters

Parameter	Description	Values	Default Value
DCS_MODE	RISING edge mode; FALLING edge mode; Clock Buffer Mode	CLK0,CLK1,CLK2,CLK3, GND,VCC,RISING,FALLIN, CLK0_GND,CLK1_GND, CLK2_GND,CLK3_GND, CLK0_VCC,CLK1_VCC, CLK2_VCC,CLK3_VCC	RISING

3.3.5 Functional Diagram

Figure 3-4 DCS Concept

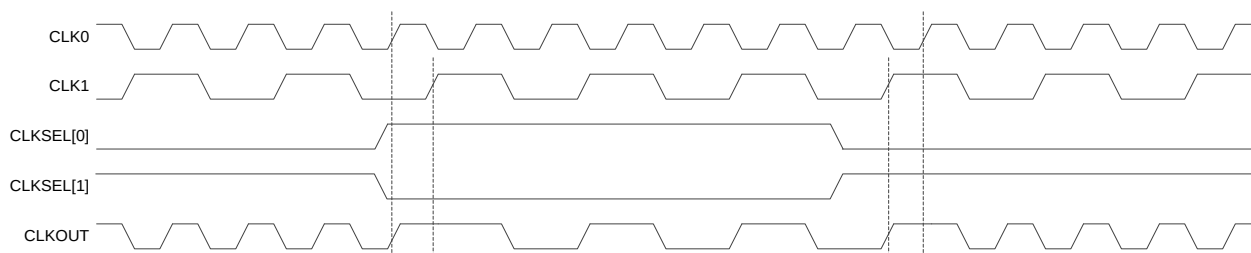


3.3.6 Timing Diagrams

1. DCS Rising Edge

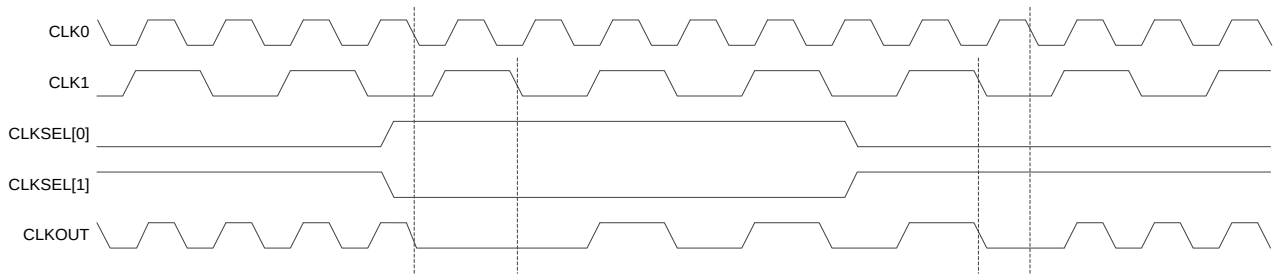
Stay as 1 after current selected clock rising edge, and the next select clock will be effective after its 1st rising edge, as shown in Figure 3-5.

Figure 3-5 DCS Rising Edge



2. DCS Falling Edge

Stay as 0 after current selected clock falling edge, and the next select clock will be effective after its first falling edge, as shown in Figure 3-6.

Figure 3-6 DCS Falling Edge

3. Clock Buffer Mode

In this mode, the DCS acts as a clock buffer.

3.4 DQCE

3.4.1 Introduction

GCLK0~GCLK5 can be turned on or off by DQCE (Dynamic Quadrant Clock Enable). When GCLK0~GCLK5 in the quadrant is off, all the logic driven by it will not toggle; therefore, lower power can be achieved.

3.4.2 Primitive

VHDL

```
COMPONENT DQCE
  PORT (
    CLKOUT : OUT std_logic;
    CE : IN std_logic;
    CLKIN : IN std_logic
  );
end COMPONENT;
end components;
```

Verilog

```
module DQCE(CLKIN,CE,CLKOUT);
  input CLKIN;
  input CE;
  output CLKOUT;
endmodule
```

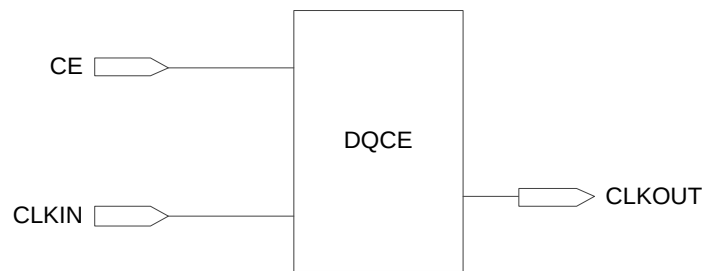
3.4.3 Port

Table 3-4 DQCE Port Signal

Port	Description
CE	Enable signal, active high.
CLKIN	Clock input
CLKOUT	Clock output

3.4.4 Functional Diagram

Figure 3-7 DQCE Concept



4HCLK

4.1 Overview

HCLK is the high-speed clock in the GOWINSEMI FPGA products. It offers low jitter and low deviation performance and, as such, can support high-speed data transfer. It is mainly suitable for source synchronous data transfer protocols.

GW1N-1 has two HCLK, located in Bank2.

GW1N-4/ GW1NR-4 have six HCLKs. Bank1, Bank2, and Bank3 all have two HCLK, with one located in the upper part of each bank, and one located in the lower part.

GW1N-9/ GW1NR-9 have eight HCLKs. Bank0, Bank1, Bank2, and Bank3 all have two HCLKs, with one located in the upper part of each bank, and one located in the lower part. HCLK input signals can be sent from bank to bank.

GW1NS-2 has 8 HCLKs. Bank0, Bank1, Bank2, and Bank3 have two HCLKs.

Figure 4-1 GW1N-4/GW1NR-4 HCLK Distribution

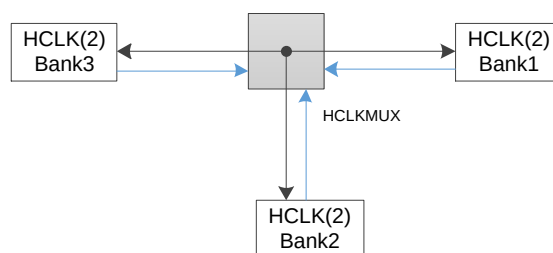
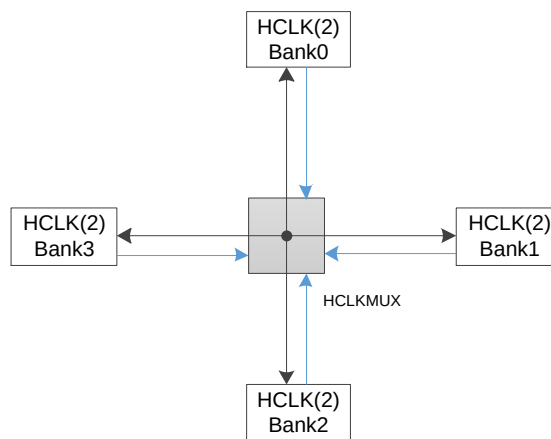
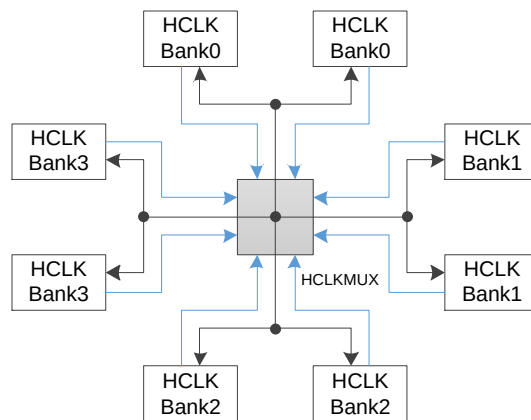
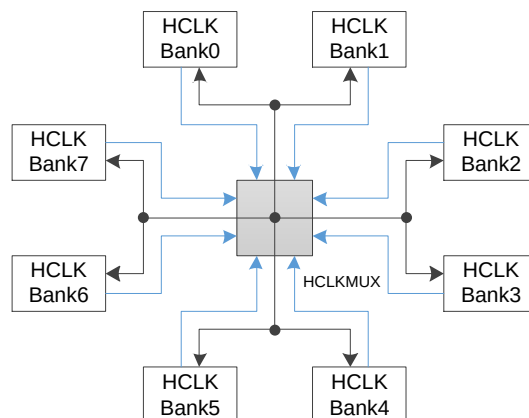


Figure 4-2 GW1N-9/GW1NR-9 HCLK Distribution**Figure 4-3 GW1NS-2 HCLK Distribution**

The GW2A/GW2AR series of products have eight HCLKs. Each bank has one HCLK.

Figure 4-4 GW2A/GW2AR series HCLK View

There is an HCLKMUX module in the middle of the high-speed clock HCLK. HCLKMUX can send the HCLK clock signal from bank to bank; as such, HCLK can be used flexibly.

Users can employ the HCLK to make use of the following function modules:

- DHCE: The dynamic high-speed clock enable module, functions in a similar way to DQCE. Users can dynamically turn on / off the high-speed clock signal;
- CLKDIV: High-speed clock divider module. Each bank has a CLKDIV that generates a clock divided by the input clock phase, which is used in the IO logic mode;
- DHCS: Dynamic high speed clock selector;
- PADCLKDEL: Dynamic delay adjustment module, produces the clock signal for the dedicated clock pin input.

4.2 CLKDIV

4.2.1 Introduction

The output of CLKDIV is the input clock divided by either 2, 3.5, 4, or 5. CLKDIV also provides the clock for the IO logic of IDES4/IVIDEO/OSER4/OVIDEO.

4.2.2 Primitive

VHDL

```

COMPONENT CLKDIV
  GENERIC(
    DIV_MODE : STRING := "2";
    GSREN : STRING := "false"
  );
  PORT(
    HCLKIN : IN std_logic;
    RESETN : IN std_logic;
    CALIB : In std_logic;
    CLKOUT : OUT std_logic
  );
end COMPONENT;
```

Verilog

```

module CLKDIV(HCLKIN, RESETN, CALIB, CLKOUT);
  input HCLKIN;
  input RESETN;
  input CALIB;
  output CLKOUT;
```

```

parameter DIV_MODE = "2";
parameter GSREN = "false";
endmodule

```

4.2.3 Port

Table 4-1 CLKDIV Port Signal

Port	Description
HCLKIN	Clock input
RESETN	Reset signal, active low
CALIB:	Dynamic signal adjustment, adjust output clock
CLKOUT	Clock output

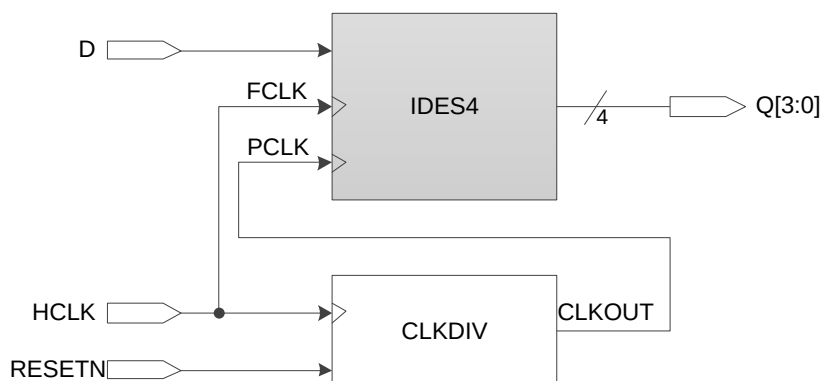
4.2.4 Parameters

Table 4-2 CLKDIV Parameters

Parameters	Description	Default Value
DIV_MODE	Division factor: 2, 3.5, 4, 5, 8	2
GSREN	Global reset enable signal: False, true	false

4.2.5 Application Schematic Diagram

Figure 4-5 CLKDIV Application Schematic Diagram



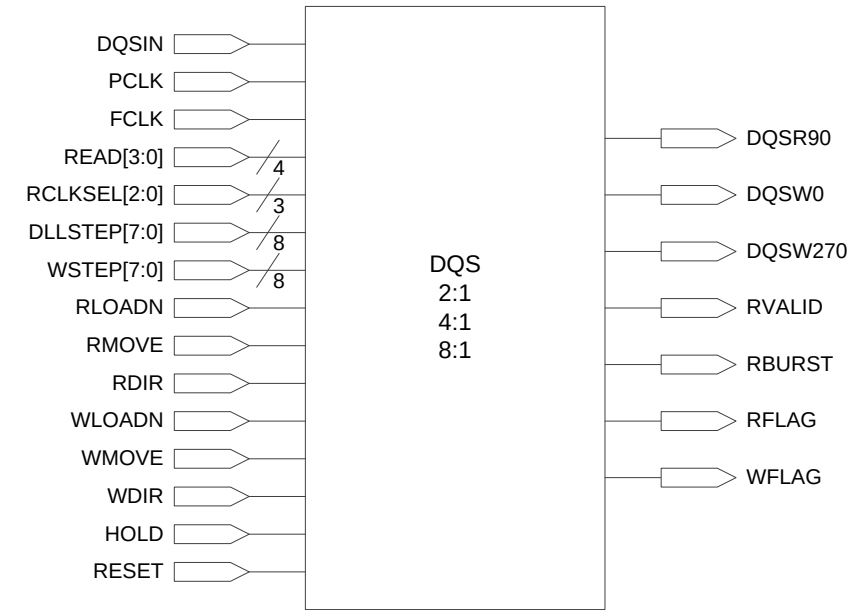
4.3 DDR Memory Interface Clock Management DQS

The DQS module of the GW2A/GW2AR series of FPGA products provides the following features to support the clock requirements of the DDR memory interface:

- Receive DQS input, sort out waveform and shift 1/4 phase;
- Provide a read / write pointer for input buffer;
- Provide valid data for internal logic;
- Provide DDR output clock signal;
- Supports DDR3 write voltage control.

The DQS module has three operating modes to meet the needs of different IO interfaces, as shown in Figure 4-6.

Figure 4-6 DQS



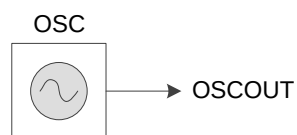
5_{osc}

5.1 Overview

GOWINSEMI FPGA products have an internal programmable oscillator that can be used as a clock frequency source. It offers low jitter and high precision. The OSC frequency is 2.1 MHz ~125 MHz, and the accuracy is $\pm 5\%$.

5.2 Functional Block Diagram

Figure 5-1 OSC View



5.3 Primitive

VHDL

```

COMPONENT OSC
  GENERIC (FREQ_DIV : integer := 128
    );
  PORT (
    OSCOUT : out std_logic
  );
end COMPONENT;
```

Verilog

```

module OSC ( OSCOUT );
  parameter  FREQ_DIV = 128;
  output OSCOUT;
endmodule
```

5.4 Port

Table 5-1 OSC Port Signal

Port	Description
OSCOUT	Clock output

5.5 Parameters

Table 5-2 OSC Parameters

Parameters	Description	Default Value
FREQ_DIV	Division factor with the range of 2~128, support even numbers only	128

5.6 Frequency Value

The on-chip oscillator also provides a clock resource for user designs. Up to 64 clock frequencies can be obtained by varying the parameters. The following formula is employed to get the output clock frequency:

$$\text{OSCOUT} = 250\text{MHz} / \text{FREQ_DIV}.$$

“FREQ_DIV” is the configuration parameter and has a range of between 2~128. It supports even numbers only.

6_{DLL}

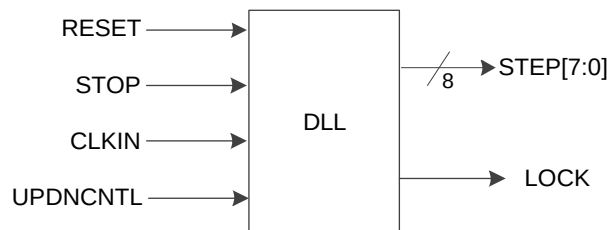
6.1 Overview

GOWINSEMI FPGA products support DLL. DLL can be used for clock reference delay. A PCLK clock that meets the delay requirement of IDDR, IDES4/8/10, and IVideo can be generated by sending the output signal STEP to the DLLDLY module.

The clock source of DLL can come from GCLKs and the neighboring HCLKs.

6.2 Functional Block Diagram

Figure 6-1 DLL View



6.3 Primitive

6.3.1 DLL Primitive

VHDL

COMPONENT DLL

GENERIC(

DLL_FORCE : integer := 0;

CLKINSEL : STRING := "00001";

DIV_SEL : bit := '1';

CODESCAL : STRING := "000";

SCAL_EN : STRING := "true"

);

PORT(

```

        CLKIN:IN std_logic:='0';
        STOP:  In std_logic:='0';
        RESET : In std_logic:='0';
        UPDNCNTL : In std_logic:='0';
        LOCK : OUT std_logic;
        STEP : OUT std_logic_vector(7 downto 0)
    );
end COMPONENT;

```

Verilog

```

module DLL (HCLKIN, CLKIN, STOP, RESET, UPDNCNTL, STEP, LOCK);
input CLKIN;
input STOP;
input UPDNCNTL;
input RESET;
output [7:0]STEP;
output LOCK;
parameter DLL_FORCE = 0;
parameterCLKINSEL="00001";
parameter CODESCAL="000";
parameter SCAL_EN="true";
parameter DIV_SEL = 1'b0;
endmodule

```

6.3.2 DLLDLY Primitive

VHDL

```

COMPONENT DLLDLY
    GENERIC(
        DLL_INSEL : bit := '0';
        DLY_SIGN : bit := '0';
        DLY_ADJ : STRING := "00000000"
    );
    PORT(
        DLLSTEP : IN std_logic_vector(7 downto 0);
        CLKIN:IN std_logic;
        DIR,LOADN,MOVE:  In std_logic;
        CLKOUT : OUT std_logic;
        FLAG : OUT std_logic
    );
end COMPONENT;

```

Verilog

```

module DLLDLY (CLKIN, DLLSTEP, DIR, LOADN, MOVE, CLKOUT, FLAG);
input CLKIN;
input [7:0] DLLSTEP;
input DIR,LOADN,MOVE;
output CLKOUT;
output FLAG;
parameter DLL_INSEL = 1'b0;
parameter DLY_SIGN = 1'b0;
parameter DLY_ADJ = "00000000";
endmodule

```

6.4 Port

6.4.1 DLL Port Signal

Table 6-1 DLL Port Signal

Port	I/O	Description
CLKIN	I	Global clock input.
STOP	I	Clock control signal: 0: Enable input clock and internal oscillator 1: Disable input clock and internal oscillator
RESET	I	DLL Reset signal: 1: Reset DLL 0: Not Reset
UPDNCNTL	I	Update control signal: 0: Update 1: Keep
STEP	O	DLL delay adjustment amount
LOCK	O	Lock the index signal: 1: Locked 0: Unlocked

6.4.2 DLLDLY Port

Table 6-2 DLLDLY Port Signal

Port	Description
DLLSTEP	Delay adjustment amount
CLKIN	Input clock
DIR	Adjust the direction, take effect at the falling edge of MOVE 1: Adjust ahead 0: Adjust backward
LOADN	Load control: 0: Load DLLSTEP data, take effect at the falling edge of MOVE 1: Invalid
MOVE	Move DLLSTEP, DIR data, take effect at the falling edge
CLKOUT	Output clock
FLAG	Overflow flag generated when the sum of the DLLSTEP and fine adjustment value exceeds 255. 1: Overflow 0: Normal

6.5 Parameters

6.5.1 DLL Parameters

Table 6-3 DLL Parameters

Parameters	Description	Default Value
DLL_FORCE	DLL force mode: 1: Forced lock, step value is "FF" 0: DLL loop lock	0
CODESCAL	000: 101° phase shift 001: 112° phase shift 010: 123° phase shift 011: 135° phase shift 100: 79° phase shift 101: 68° phase shift 110: 57° phase shift 111: 45° phase shift	000
SCAL_EN	SCAL enable signal: true: enable CODESCAL false: invalid CODESCAL, constant output of 90° phase shift	true
DIV_SEL	Lcok mode: 1: Fast locking 0: Normal locking	0

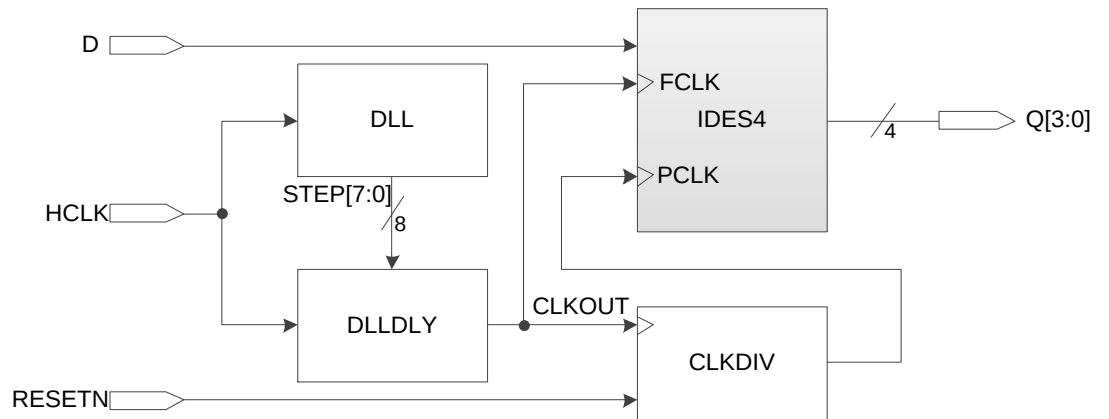
6.5.2 DLLDLY Parameter

Table 6-4 DLLDLY Parameter

Parameters	Description	Default Value
DLL_INSEL	Bypass control: 0: Bypass DLL_DLY 1: Start DLL_DLY	0
DLY_SIGN	Fine adjustment sign bit: 0: "+" 1: "-"	0
DLY_ADJ	Fine adjustment value: When DLY_SIGN=0 00000000: +0 00000001: +1 11111111: +255 When DLY_SIGN=1 00000000: -256 00000001: -255 11111111: -1	00000000

6.6 Application Schematic Diagram

Figure 6-2 DLL Application Schematic Diagram



7 PLL

7.1 Overview

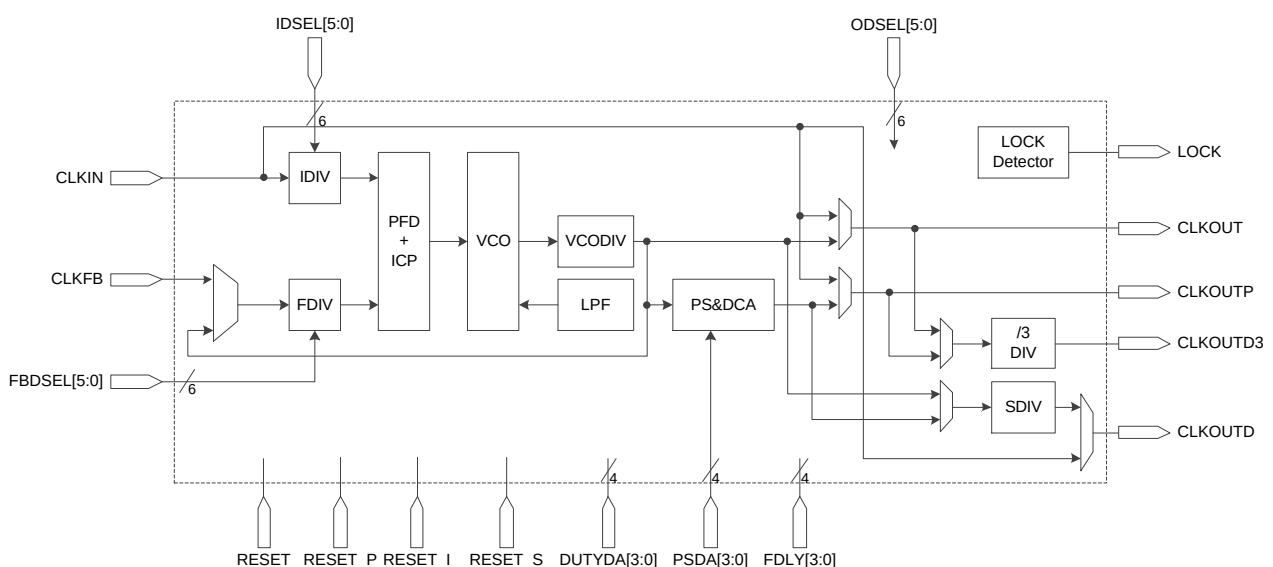
PLL (Phase-locked Loop) is one kind of feedback control circuit. The frequency and phase of the internal oscillator signal is controlled by the external input reference clock.

PLL blocks provide the ability to synthesize clock frequencies. Frequency adjustment (multiply and division), phase adjustment, and duty cycle can be adjusted by configuring the parameters.

The PLL reference clock source can come from an external PLL pin or from internal routing GCLK, HCLK, or general data signal. The PLL feedback signal can come from the external PLL feedback input or from internal routing GCLK, HCLK, or general data signal.

7.2 Functional Block Diagram

Figure 7-1 PLL Structure



7.3 Primitive

VHDL

COMPONENT PLL

GENERIC(

```

    FCLKIN : STRING := "100.0"; --frequency of the clkin(M)
    DEVICE : STRING := "GW1N2K";
    --"GW1N1K","GW1N2K","GW1N4K","GW1N6K","GW1N9K"
    DYN_IDIV_SEL : STRING := "false";
    --true:IDSEL; false:IDIV_SEL
    IDIV_SEL : integer := 0;
    --Input divider IDIV, 0:1,1:2...63:64. 1~64
    DYN_FBDIV_SEL : STRING := "false";
    FBDIV_SEL : integer := 0;
    --Feedback divider FBDIV, 0:1,1:2...63:64. 1~64
    DYN_ODIV_SEL : STRING := "false";
    --true:ODSEL; false:ODIV_SEL
    ODIV_SEL : integer := 8;--2/4/8/16/32/48/64/80/96/112/128
    PSDA_SEL : STRING := "0000";
    DYN_DA_EN : STRING := "false";
    --true:PSDA or DUTYDA or FDA; false: DA_SEL
    DUTYDA_SEL : STRING := "1000";
    CLKOUT_FT_DIR : bit := '1';
    -- CLKOUT fine tuning direction. '1' only
    CLKOUTP_FT_DIR : bit := '1'; -- '1' only
    CLKOUT_DLY_STEP : integer := 0; -- 0,1,2,4
    CLKOUTP_DLY_STEP : integer := 0; -- 0,1,2
    CLKOUTD3_SRC : STRING := "CLKOUT";
    --select div3 output, CLKOUTP or CLKOUT
    CLKFB_SEL : STRING := "internal";
    CLKOUT_BYPASS : STRING := "false";
    CLKOUTP_BYPASS : STRING := "false";
    CLKOUTD_BYPASS : STRING := "false";
    CLKOUTD_SRC : STRING := "CLKOUT";
    --select div output, CLKOUTP or CLKOUT
    DYN_SDIV_SEL : integer := 2 -- 2~128,only even num

```

);

PORT(

```

    CLKIN      : in std_logic_vector;

```

```

        CLKFB    : in std_logic;
        ODSEL     : in std_logic_vector(5 downto 0);
        IDSEL     : in std_logic_vector(5 downto 0);
        FBDSEL    : in std_logic_vector(5 downto 0);
        RESET     : in std_logic: ='0';
        RESET_P   : in std_logic: ='0';
        RESET_I   : in std_logic: ='0';
        RESET_S   : in std_logic : ='0';
        PSDA, FDLY: in std_logic_vector(3 downto 0);
        DUTYDA    : in std_logic_vector(3 downto 0);
        LOCK      : out std_logic;
        CLKOUT     : out std_logic;
        CLKOUTD    : out std_logic;
        CLKOUTP    : out std_logic;
        CLKOUTD3:  out std_logic
    );
end COMPONENT;

```

Verilog

```

module PLL (CLKOUT, CLKOUTP, CLKOUTD, CLKOUTD3, LOCK, CLKIN,
            CLKFB, FBDSEL, IDSEL, ODSEL, DUTYDA, PSDA, FDLY, RESET, RESET_P,
            RESET_I, RESET_S);
    input CLKIN;
    input CLKFB;
    input RESET;
    input RESET_P;
    input RESET_I;
    input RESET_S;
    input [5:0] ODSEL;
    input [5:0] FBDSEL;
    input [5:0] IDSEL;
    input [3:0] PSDA,FDLY;
    input [3:0] DUTYDA;
    output CLKOUT;
    output LOCK;
    output CLKOUTP;
    output CLKOUTD;
    output CLKOUTD3;
    parameter FCLKIN = "100.0"; // frequency of the clkIn(M)
    parameter DYN_IDIV_SEL= "false";//true:IDSEL; false:IDIV_SEL

```

```

parameter IDIV_SEL = 0; // 0:1,1:2...63:64. 1~64
parameter DYN_FBDIV_SEL= "false";
parameter FBDIV_SEL = 0; // 0:1,1:2...63:64. 1~64
parameter DYN_ODIV_SEL= "false";//true:ODSEL; false:ODIV_SEL
parameter ODIV_SEL = 8; // 2/4/8/16/32/48/64/80/96/112/128
parameter PSDA_SEL= "0000";//
parameter DYN_DA_EN = "false";
//true:PSDA or DUTYDA or FDA; false:  DA_SEL
parameter DUTYDA_SEL= "1000";//
parameter CLKOUT_FT_DIR = 1'b1; // CLKOUT fine tuning direction. 1'b1 only
parameter CLKOUTP_FT_DIR = 1'b1; // 1'b1 only
parameter CLKOUT_DLY_STEP = 0; // 0,1,2,4
parameter CLKOUTP_DLY_STEP = 0; // 0,1,2
parameter CLKFB_SEL = "internal";//"internal","external";
parameter CLKOUT_BYPASS = "false"; // "true"; "false"
parameter CLKOUTP_BYPASS = "false"; // "true"; "false"
parameter CLKOUTD_BYPASS = "false"; // "true"; "false"
parameter DYN_SDIV_SEL = 2; // 2~128,only even num
parameter CLKOUTD_SRC = "CLKOUT"; //CLKOUT,CLKOUTP
parameter CLKOUTD3_SRC = "CLKOUT"; //CLKOUT,CLKOUTP
parameter DEVICE = "GW1N2K";
// "GW1N1K","GW1N2K","GW1N4K","GW1N6K","GW1N9K"
endmodule

```

7.4 Port Signal

Table 7-1 PLL Port Signal

Port Name	I/O	Description
CLKIN	I	Reference clock input
CLKFB	I	Feedback clock input
RESET	I	PLL reset
RESET_P	I	PLL power down
RESET_I	I	IDIV reset
RESET_S	I	SDIV and DIV3 reset
ODSEL[5: 0]	I	Dynamic ODIV control: 2, 4, 8, 16, 32, 48, 64, 80, 96, 112, and 128.
IDSEL [5: 0]	I	Dynamic IDIV control: 1~64
FBDSEL [5: 0]	I	Dynamic FDIV control: 1~64
PSDA [3: 0]	I	Dynamic phase control (take effect at rising edge)
DUTYDA [3: 0]	I	Dynamic duty cycle control (take effect at falling edge)

Port Name	I/O	Description
FDLY[3:0]	I	CLKOUTP dynamic delay control
CLKOUT	O	Clock output with no phase and duty cycle adjustment
CLKOUTP	O	Clock output with phase and duty cycle adjustment
CLKOUTD	O	Clock divider from CLKOUT and CLKOUTP (controlled by SDIV)
CLKOUTD3	O	clock divider from CLKOUT and CLKOUTP (controlled by DIV3 with the constant division value 3)
LOCK	O	PLL lock status: 1: Locked 0: Unlocked

Table 7-2 Comparison of IDSEL Port Parameters

IDSEL[5: 0]	IDIV Parameter Value
111111	1
111110	2
111101	3
111100	4
111011	5
111010	6
111001	7
111000	8
110111	9
.....	
000000	64

Table 7-3 Comparison of FBDSEL Port Parameters

FBDSEL [5: 0]	FDIV Parameter Value
111111	1
111110	2
111101	3
111100	4
111011	5
111010	6
111001	7
111000	8
110111	9
.....	
000000	64

Table 7-4 Comparison of ODSEL Port Parameters

ODSEL [5: 0]	ODIV Parameter Value
111111	2
111110	4
111100	8
111000	16
110000	32
101000	48
100000	64
011000	80
010000	96
001000	112
000000	128

7.5 Parameters

Table 7-5 PLL Parameters

Parameters	Description
FCLKIN	CLKIN frequency value, STRING=100.0 (the default value is 100MHz);
DYN_ODIV_SEL	STRING=false; false: Use parameters to input ODIV_SEL value; true: Use parameters or port to input ODSEL value.
DEVICE	STRING=GW1N2K; "GW1N1K","GW1N2K","GW1N4K","GW1N6K","GW1N9K", "GW2A18","GW2A55"

Parameters	Description
DYN_IDIV_SEL	STRING=false; false: Use parameters to input IN_SEL value; true: Use port to input INSEL value.
IDIV_SEL	Type: Integer=0; The value of IDIV_SEL input range is 0~63; IDIV=IDIV_SEL+1. The range is 1~64.
DYN_FBDIV_SEL	STRING=false; false: Use parameters to input FBDIV_SEL value; true: Use port to input FBDSEL value.
FBDIV_SEL	Type: Integer=0; The value range of FBDIV_SEL is 0 ~ 63; FDIV=FBDIV_SEL+1. The range is 1 ~ 64.
DYN_SDIV_SEL	Type: integer=2; The value range is 2~128. Input even numbers only.
ODIV_SEL	ODIV value, Type: Integer=8; Optional value: 2, 4, 8, 16, 32, 48, 64, 80, 96, 112, 128.
PSDA_SEL	Phase adjustment parameters setting, STRING = 0000; The value range is 0000~1111 (The default value is 0000).
DUTYDA_SEL	Duty cycle adjustment parameters setting, STRING = 0000; The value range is 0000~1111 (The default value is 0000).
DYN_DA_EN	STRING=false; false: Select the signal PSDA_SEL, DUTYDA_SEL as the signal of phase duty cycle adjustment; true: Select the signal PSDA, DUTYDA, FDLY, etc. as the control signal of phase duty cycle adjustment.
CLKOUT_FT_DIR	bit =1; CLKOUT fine adjustment direction: 0:+ 1:-
CLKOUT_DLY_STEP	integer= 0, the value is 0, 1, 2, 4; CLKOUT fine adjustment value: parameter value x delay(delay=50ps).
CLKOUTP_FT_DIR	CLKOUTP fine adjustment direction: 1'b1:- 1'b0:+
CLKOUTP_DLY_STEP	integer= 0, the value is 0, 1, 2; CLKOUT fine adjustment value: parameter value x delay (delay=50ps).
CLKFB_SEL	STRING=GW1N2K; CLKOUT: Select internal feedback; CLKFB: Select the external feedback of CLKFB.
CLKOUTD_SRC	CLKOUTD source signal, STRING =CLKOUT; CLKOUT: Select CLKOUT; CLKOUTP: Select CLKOUTP;
CLKOUTD3_SRC	CLKOUTD3 source signal, STRING =CLKOUT; CLKOUT: Select CLKOUT; CLKOUTP: Select CLKOUTP;
CLKOUT_BYPASS	CLKOUT bypass control, STRING =false; true: Bypass PLL, CLKOUT= CLKIN; false: Enable CLKOUT output
CLKOUTP_BYPASS	CLKOUTP bypass control, STRING =false; true: bypass PLL, CLKOUT= CLKIN; false: enable CLKOUTP output
CLKOUTD_BYPASS	CLKOUTD bypass control, STRING =false; true: Bypass PLL, CLKOUT= CLKIN;

Parameters	Description
	false: Enable CLKOUTD output.

Table 7-6 PLL Phase Parameters Adjustment

ParameterPSDA_SEL or Port PSDASetting	Phase Adjustment
0000	0°
0001	22.5°
0010	45°
0011	67.5°
0100	90°
0101	112.5°
0110	135°
0111	157.5°
1000	180°
1001	202.5°
1010	225°
1011	247.5°
1100	270°
1101	292.5°
1110	315°
1111	337.5°

Table 7-7 Adjustment of PLL Phase Parameters

ParametersDUTYDA_SEL Setting	Duty Cycle Setting (/16)
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7
1000	8
1001	9
1010	10
1011	11
1100	12
1101	13
1110	14

Duty cycle adjustment needs refer to phase shift setting. For example, when the phase shift setting is “0” (00000), the 50% duty cycle setting will be “8” (1000). If the phase shift setting is “180°”, the 50% duty cycle setting will be “0” (0000).

Dynamic phase duty cycle setting:

- If DUTYDA [3:0] > PSDA [3:0], DutyCycle=1/16 x (DUTYDA [3:0]- PSDA [3:0]).
- If DUTYDA [3:0] < PSDA [3:0], DutyCycle=1/16 x (16+ DUTYDA [3:0]- PSDA [3:0]).

Note!

The difference between DUTYDA and PSDA does not support the following three situations: (DUTYDA - PSDA = 0), (DUTYDA - PSDA = 1), and (DUTYDA - PSDA = 15).

Use FDLY[3:0] to dynamically control CLKOUTP delay. Each step increases a fix *0.125ns for total 15 steps (1.875ns). Lagging (clock signal (CLKOUTP) lags input signal) and leading (clock signal (CLKOUTP) leads the input signal) need to combine phase shift setting.

Table 7-8 PLL: Adjustment of the Delay Parameters

Port FDLY [3:0]	Delay Steps
0111	7
1011	11
1101	13
1110	14
1111	15

7.6 Frequency Calculation

PLL can adjust the frequency of the input clock CLKIN (multiply and division). The formulas for doing so are as following:

- a). $F_{CLKOUT} = (F_{CLKIN} \times F_{DIV})/IDIV$
- b). $F_{VCO} = F_{CLKOUT} \times ODIV$
- c). $F_{CLKOUTD} = F_{CLKOUT}/SDIV$
- d). $F_{PDF} = F_{CLKIN}/IDIV = F_{CLKOUT}/F_{DIV}$

Note!

- fCLKIN: the frequency of input clock CLKIN;
- fCLKOUT: the clock frequency of CLKOUT and CLKOUTP
- FCLKOUTD: the clock frequency of CLKOUTD.
- FPDF: the phase detector frequency.

Adjust IDIV, FDIV, ODIV, and SDIV to fine-tune the clock to the expected frequency.

7.7 Example

To achieve a 1.75 frequency multiplication of the input clock, set as follows: IDIV_SEL=1 (division factor is 2), FBDIV_SEL=6 (FB division factor is 7), DYN_SDIV_SEL=2 (SDIV division factor is 2). According to the division frequency equation, CLKOUT frequency is 3.5 times that of CLKIN, and CLKOUTD frequency is 1.75 times that of CLKIN.

Verilog module call:

```
.PLL#(
.FCLKIN("100.0"),
.DYN_IN_SEL("false"),
.IN_SEL(5),
.DYN_IDIV_SEL("false"),
.IDIV_SEL(1),
.DYN_FBDIV_SEL("false"),
.FBDIV_SEL(6),
.ODIV_SEL(8),
.PSDA_SEL (4b"0000"),
.DYN_DA_EN ("false"),
.DUTYDA_SEL(4b"1000"),
.CLKOUT_FT_DIR(1b'1'),
.CLKOUTP_FT_DIR(1b'1'),
.CLKOUT_DLY_STEP(0),
.CLKOUTP_DLY_STEP(0),
.CLKOUTD3_SRC("CLKOUT"),
.CLKFB_SEL(CLKOUT),
```

```
.CLKOUT_BYPASS("false"),  
.CLKOUTP_BYPASS("false"),  
.CLKOUTD_BYPASS("false"),  
.CLKOUTD_SRC(CLKOUT),  
.DYN_SDIV_SEL(2)  
  
)  
  
PLL_inst(  
.CLKIN(CLKIN),  
.CLKFB(CLKFB),  
.INSEL(INSEL),  
.IDSEL(IDSEL),  
.FBDSEL(FBDSEL),  
.RESET(RESET),  
.RESET_P(RESET_P),  
.RESET_I(RESET_I),  
.RESET_S(RESET_S),  
.PSDA(PSDA),  
.FDLY(FDLY),  
.DUTYDA(DUTYDA),  
.LOCK(LOCK),  
.CLKOUT(CLKOUT),  
.CLKOUTD(CLKOUTD),  
.CLKOUTP(CLKOUTP),  
.CLKOUTD3(CLKOUTD3);
```

